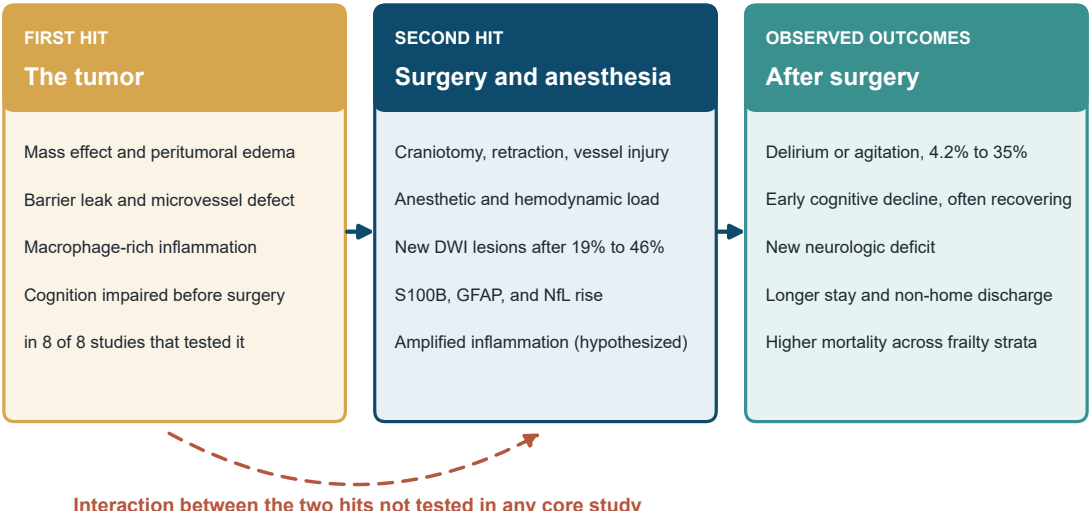
FIRST HIT
The tumor
Mass effect and peritumoral edema
Barrier leak and microvessel defect
Macrophage-rich inflammation
Cognition impaired before surgery
in 8 of 8 studies that tested it
SECOND HIT
Surgery and anesthesia
Craniotomy, retraction, vessel injury
Anesthetic and hemodynamic load
New DWI lesions after 19% to 46%
S100B, GFAP, and NfL rise
Amplified inflammation (hypothesized)
OBSERVED OUTCOMES
After surgery
Delirium or agitation, 4.2% to 35%
Early cognitive decline, often recovering
New neurologic deficit
Longer stay and non-home discharge
Higher mortality across frailty strata
Interaction between the two hits not tested in any core study